Auxiliary input

文件: aux_input.kicad_sch

Driver supply

文件: supply_12v.kicad_sch

Logic and sensors

文件: supply_logic.kicad_sch

Bus and brake

文件: bus_brake.kicad_sch

U half bridge

文件: bridge_u.kicad_sch

V half bridge

文件: bridge_v.kicad_sch

W half bridge

文件: bridge_w.kicad_sch

Current sensing

文件: current_sense.kicad_sch

Voltage and NTC

文件: voltage_sense.kicad_sch

Analog protection

文件: comparators.kicad_sch

Hardware latch

文件: fault_logic.kicad_sch

Hardware PWM gates

文件: control_interface.kicad_sch

Base interlock

文件: base_interlock.kicad_sch

Base LEDs

文件: base_leds.kicad_sch

GPIO command coupling

文件: mcu_coupling.kicad_sch

ADC and status returns

文件: core_signal_protection.kicad_sch

Encoders and SPI

文件: sensors.kicad_sch

SWD UART CAN and status

文件: debug_comms.kicad_sch

64-pin core sockets

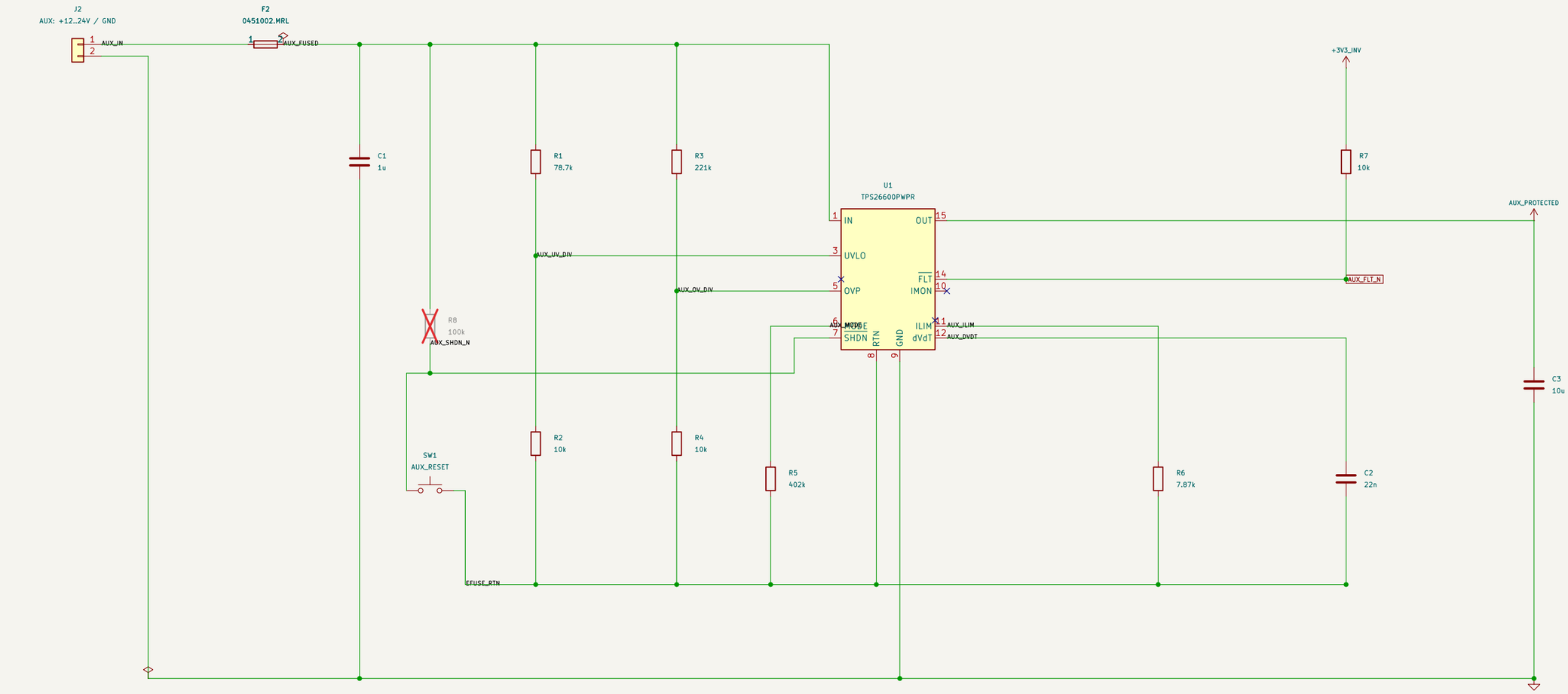
文件: power_interface.kicad_sch

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB
power_system
Sheet: /
File: inverter.kicad_sch

Title: Inverter base with GPIO minimum-system core

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 1/20

AUXILIARY INPUT / eFUSE / 12–24 V



RTN / exposed pad is the eFuse reference island. It is NOT the board GND pin.
MODE=402k selects latch-off. AUX_RESET resets the auxiliary path; no MCU startup dependency.
UV / OV / current-limit values remain unchanged; dynamics and surge coordination still require review.

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

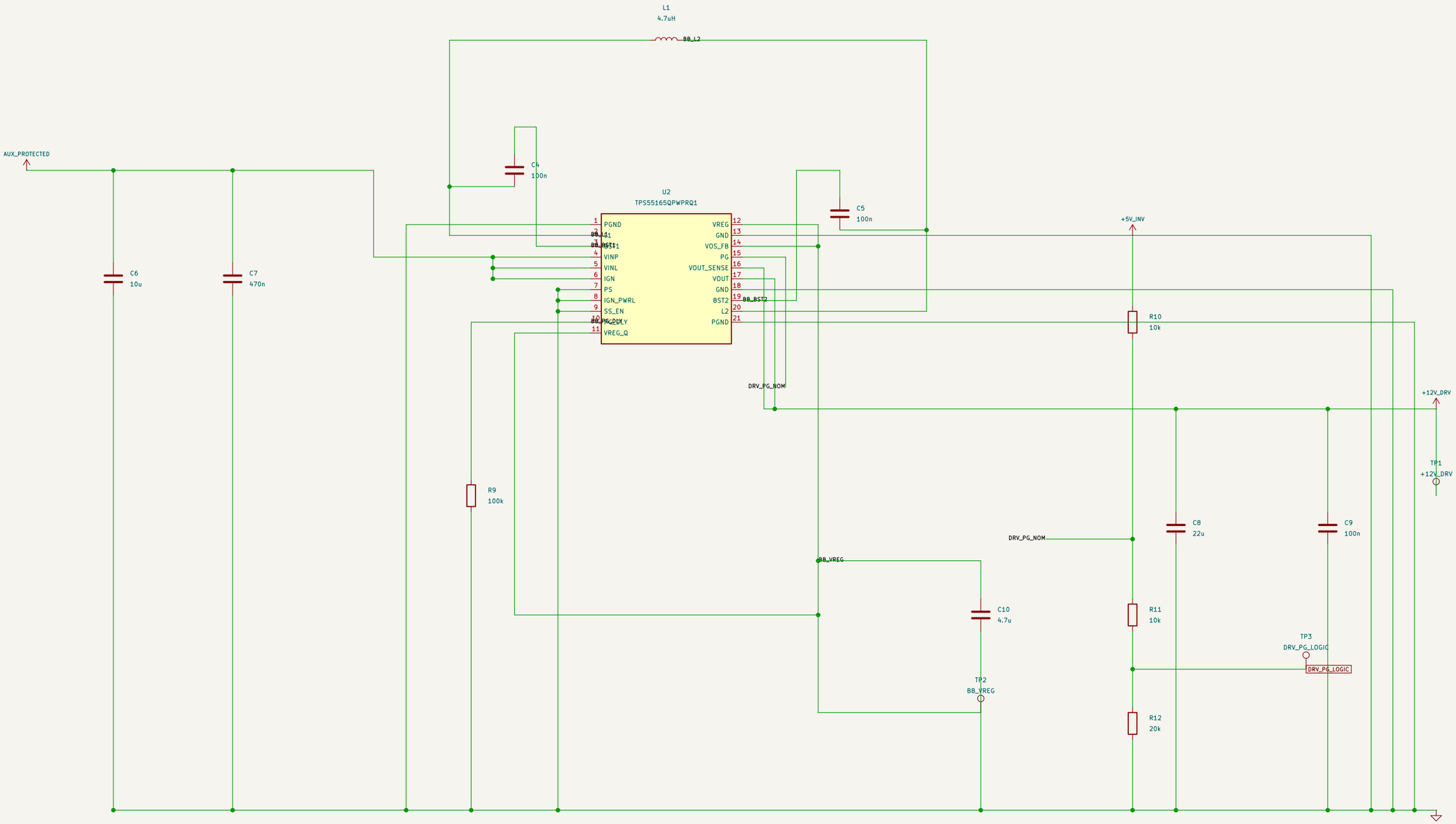
power_system

Sheet: /Auxiliary input/
File: aux_input.kicad_sch

Title: Inverter – 12–24 V auxiliary input protection

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 2/20

12 V DRIVER RAIL / TPS55165 BUCK-BOOST



VREG_Q and VOS_FB return to VREG; VREG is not a load rail. VOUT_SENSE is tied to the 12 V output.
Input/output/bootstrap wiring is shown directly. PG high is about 2.5 V with the fitted pullup/divider.
Inductor model, startup, worst-case gate voltage and thermal capability remain review items.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

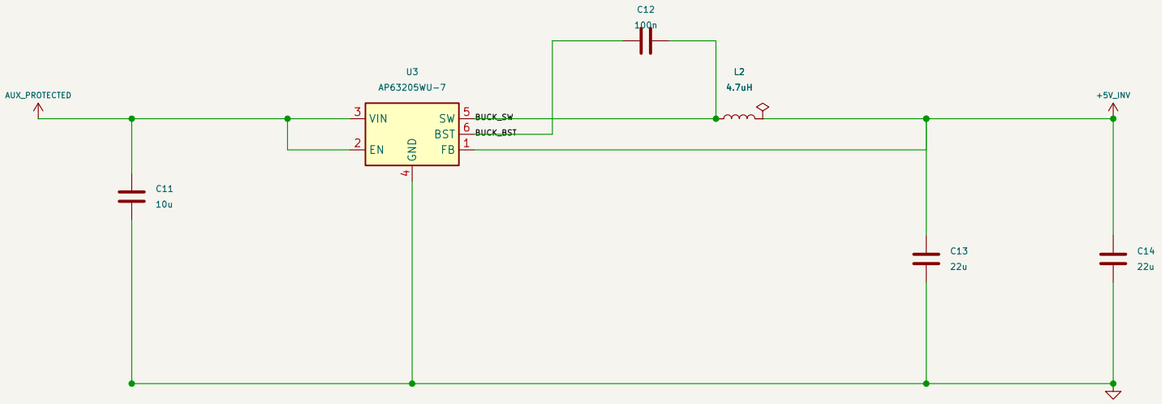
power_system

Sheet: /Driver supply/
File: supply_12v.kicad_sch

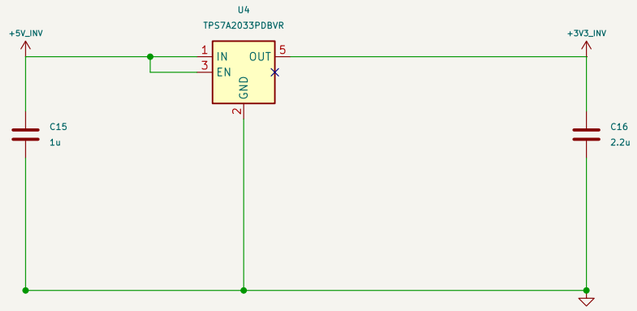
Title: Inverter - regulated 12 V driver supply

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 3/20

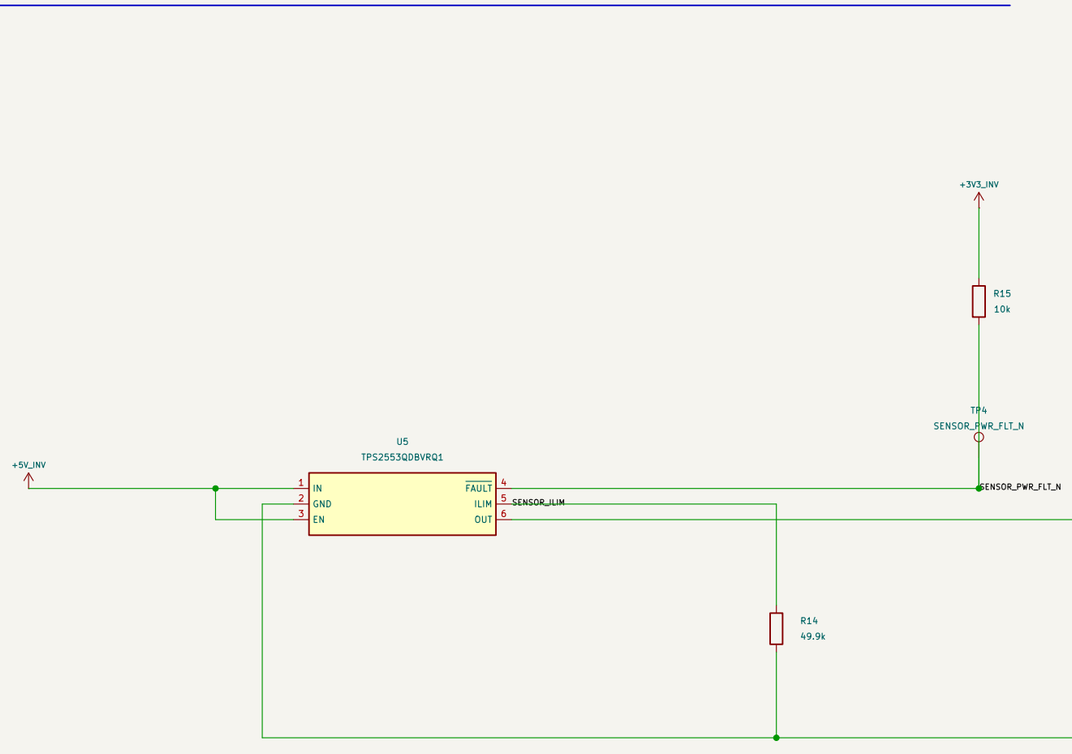
5 V BUCK / AP63205



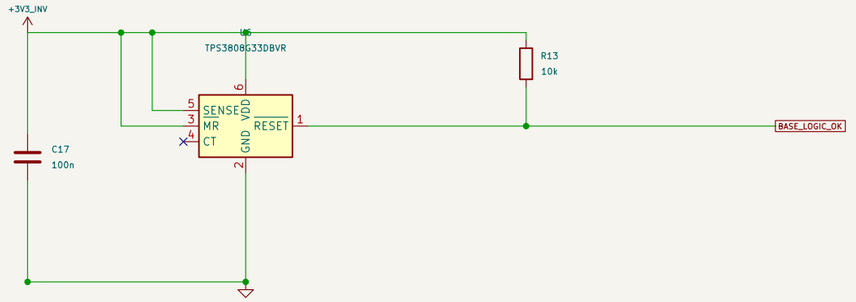
3.3 V BASE LOGIC / ANALOG SUPPLY



PROTECTED OPTIONAL SENSOR POWER / TOTAL TARGET 0.5 A



3.3 V SUPERVISOR / LOCAL BASE PERMISSION



J7 / external power links default DNP. The branch current target is total, not per connector.
U4 supplies only base analog and safety logic. BASE_LOGIC_OK never connects to controller NRST.
The removable MCU board regulates its own 3.3 V; local current budget and temperature require verification.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

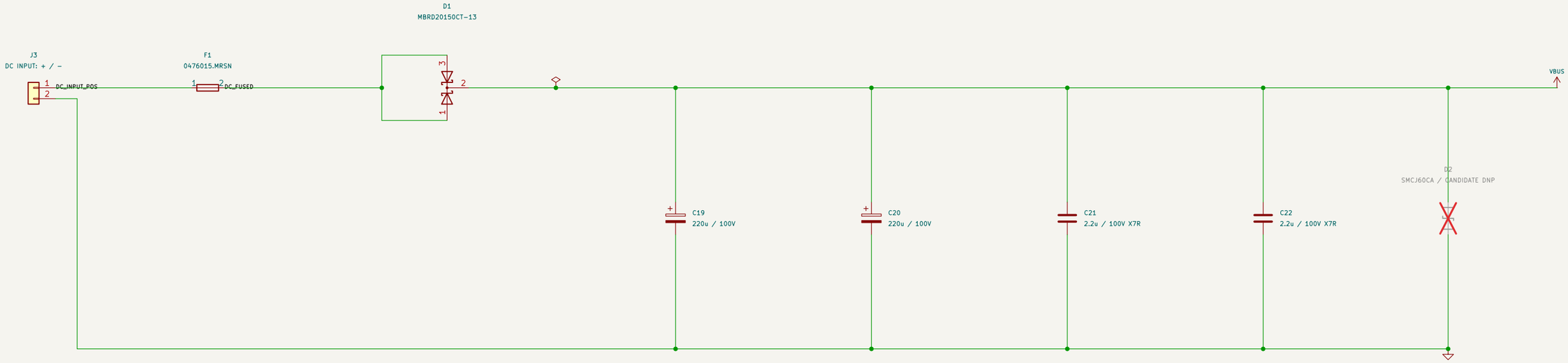
power_system

Sheet: /Logic and sensors/
File: supply_logic.kicad_sch

Title: Inverter - 5 V, 3.3 V and sensor power

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 4/20

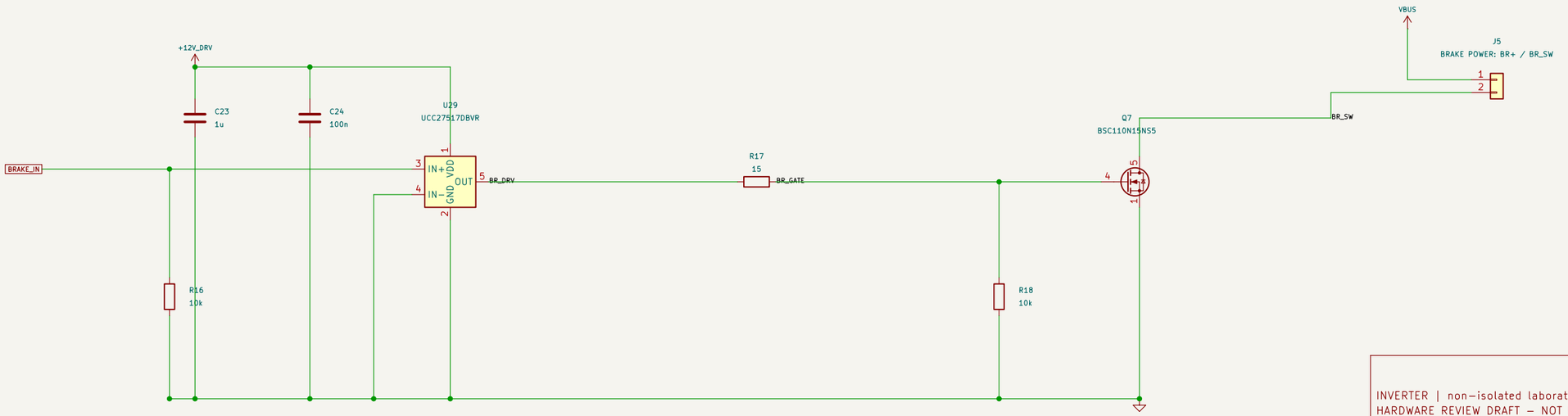
MAIN BUS INPUT / BULK CAPACITORS / RAW PWM OUTPUTS



D2 is DNP. The input diode / TVS / fuse are still procurement and energy/thermal review items.
No main-bus precharge or dedicated bleeder. U/V/W are raw bridge PWM, not LC-filtered sine outputs.



ONE BRAKE BRANCH / EXTERNAL RESISTOR / CONTROLLER HEADER COMMAND



J5: BR+ = VBUS, BR_SW = MOS drain. External resistor goes across these pins; unused port stays OPEN.
Brake is independent of the bridge ENABLE/fault gates. The base-side input pull-down defaults a missing controller command LOW.

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

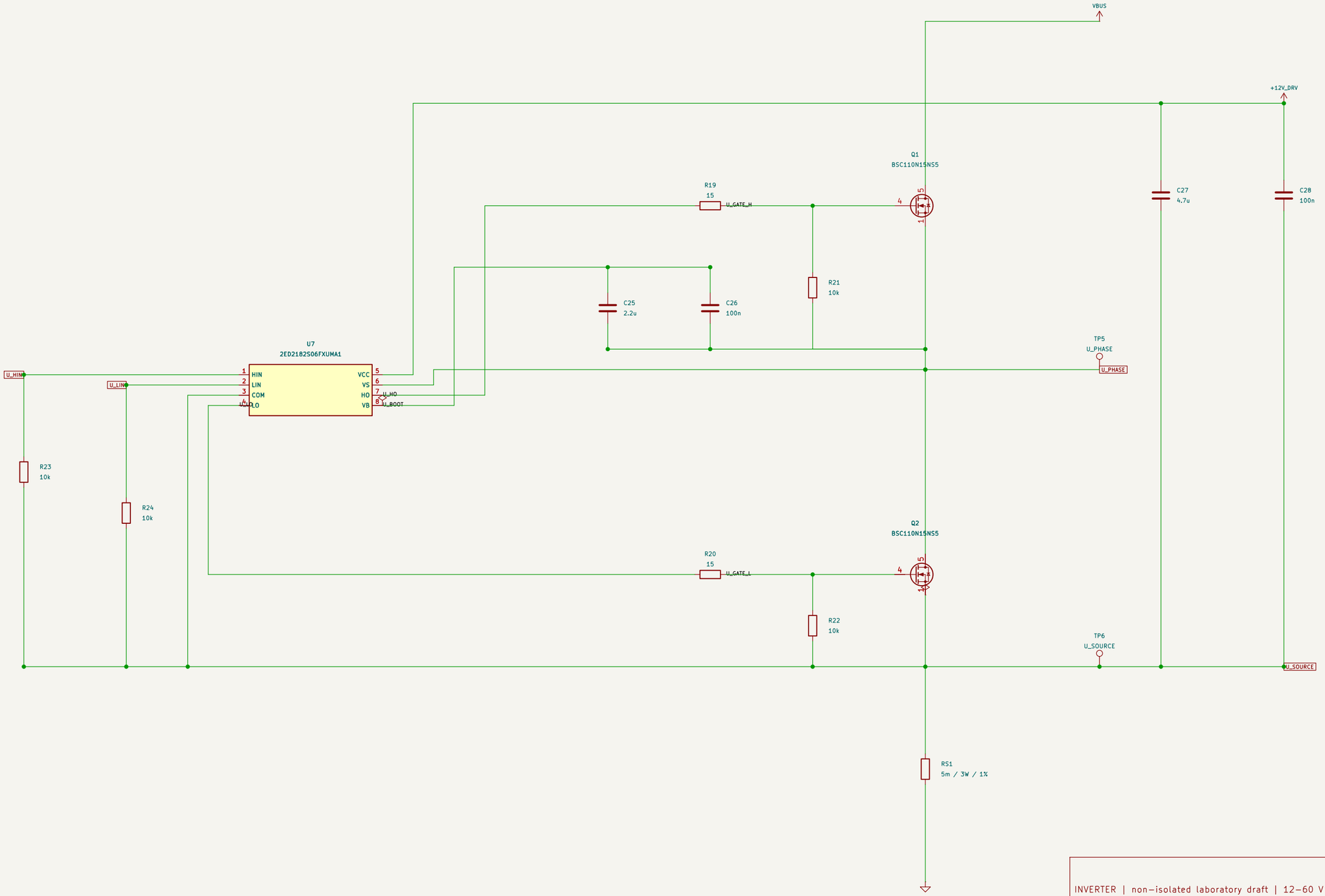
power_system

Sheet: /Bus and brake/
File: bus_brake.kicad_sch

Title: Inverter – DC bus input, bulk capacitors and brake

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 5/20

U HALF BRIDGE / VISIBLE GATE, BOOTSTRAP AND SHUNT RETURNS



COM, low-gate pulldown, local VCC capacitors and sense high all return to SOURCE above the shunt.
High-gate pulldown and bootstrap capacitors return to PHASE. Common GND/DC- exists only below RS.
Gate R / bootstrap values remain initial values; deadtime, minimum recharge time and waveforms require validation.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

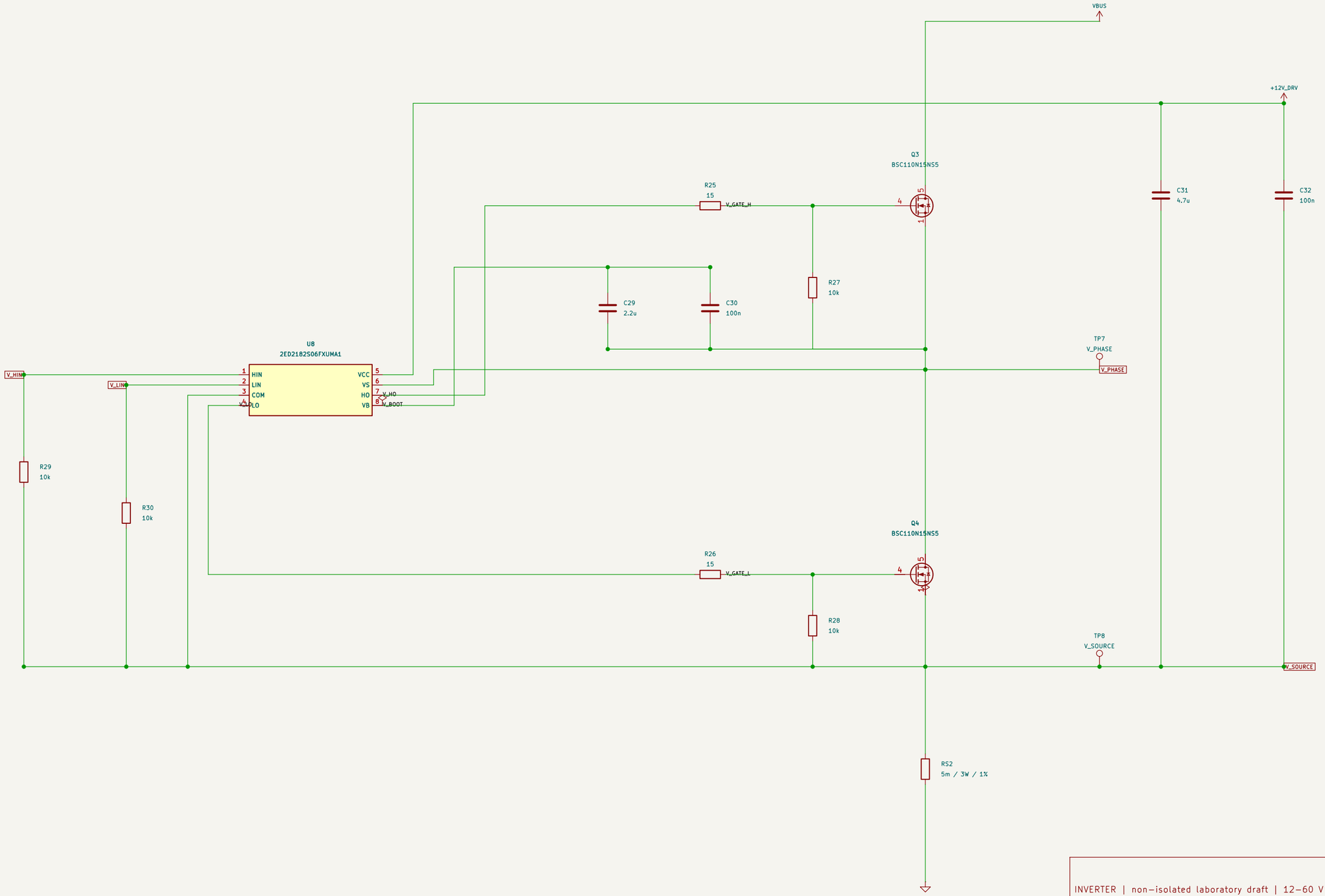
power_system

Sheet: /U half bridge/
File: bridge_u.kicad_sch

Title: Inverter - U half bridge and low-side shunt

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 6/20

V HALF BRIDGE / VISIBLE GATE, BOOTSTRAP AND SHUNT RETURNS



COM, low-gate pulldown, local VCC capacitors and sense high all return to SOURCE above the shunt.
High-gate pulldown and bootstrap capacitors return to PHASE. Common GND/DC- exists only below RS.
Gate R / bootstrap values remain initial values; deadtime, minimum recharge time and waveforms require validation.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

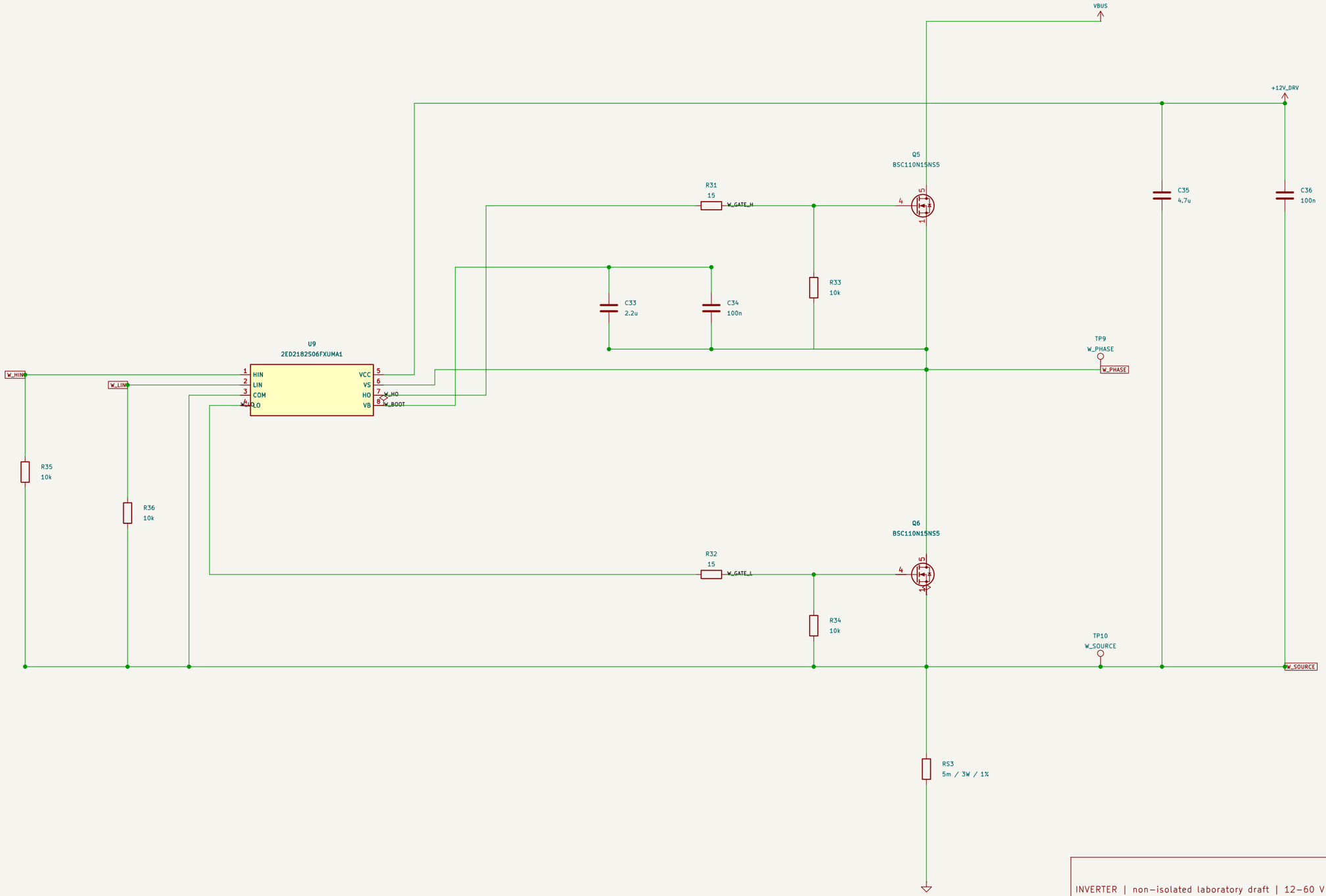
power_system

Sheet: /V half bridge/
File: bridge_v.kicad_sch

Title: Inverter - V half bridge and low-side shunt

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 7/20

W HALF BRIDGE / VISIBLE GATE, BOOTSTRAP AND SHUNT RETURNS



COM, low-gate pulldown, local VCC capacitors and sense high all return to SOURCE above the shunt.
High-gate pulldown and bootstrap capacitors return to PHASE. Common GND/DC- exists only below RS.
Gate R / bootstrap values remain initial values; deadtime, minimum recharge time and waveforms require validation.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

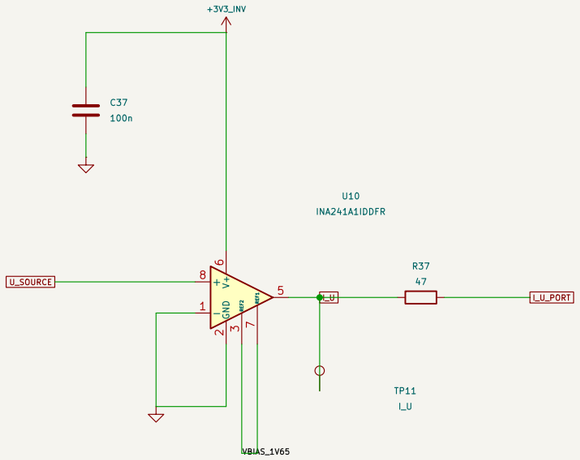
power_system

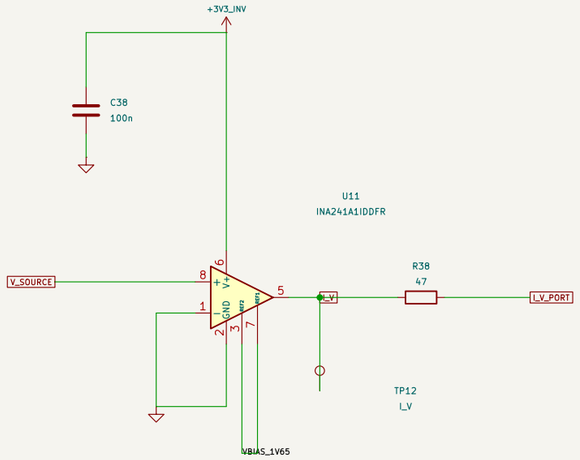
Sheet: /W half bridge/
File: bridge_w.kicad_sch

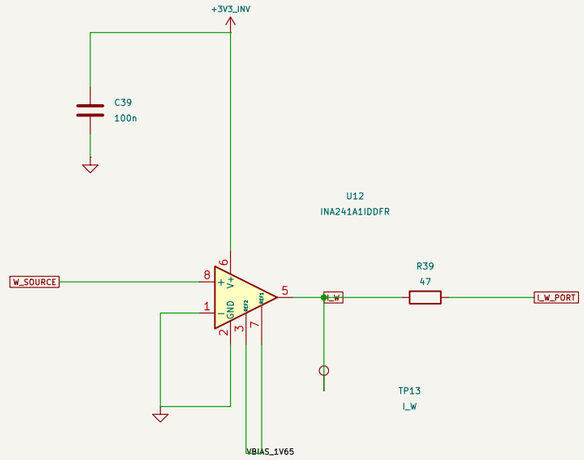
Title: Inverter - W half bridge and low-side shunt

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 8/20

THREE LOW-SIDE KELVIN CURRENT CHANNELS

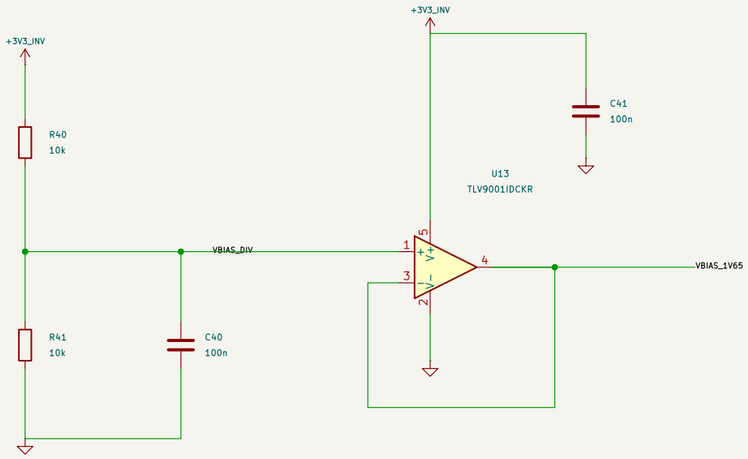


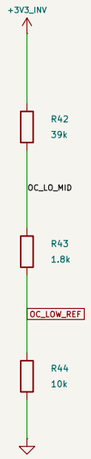


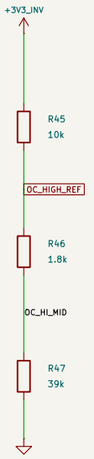


Each channel uses the buffered VBIAS_1V65 generated below.

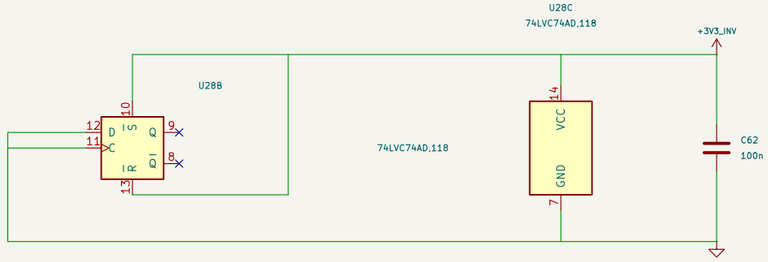
BUFFERED BIAS AND CURRENT WINDOW REFERENCES







SPARE U28B / FIXED INPUTS



I = 1.65 V + 50 mV/A; nominal current trips +/−20 A. Low-side shunts have switching-state blind intervals.
References track the local 3.3 V rail; phase-current sign and sampling windows require bench validation.

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB

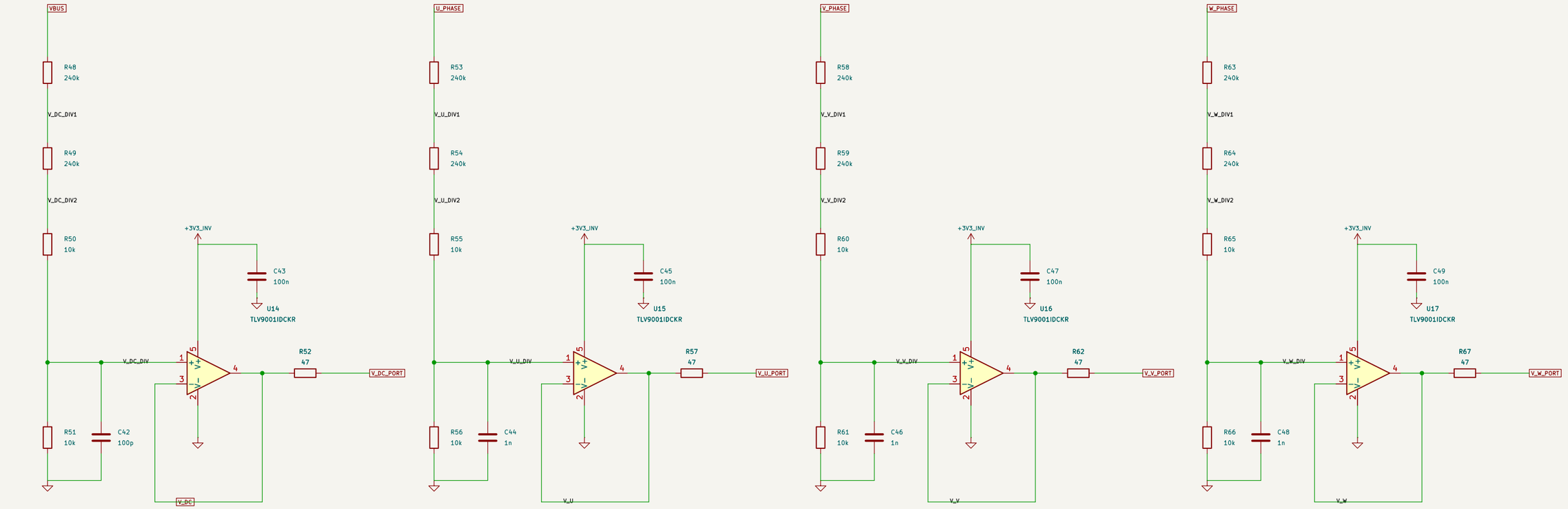
power_system

Sheet: /Current sensing/
File: current_sense.kicad_sch

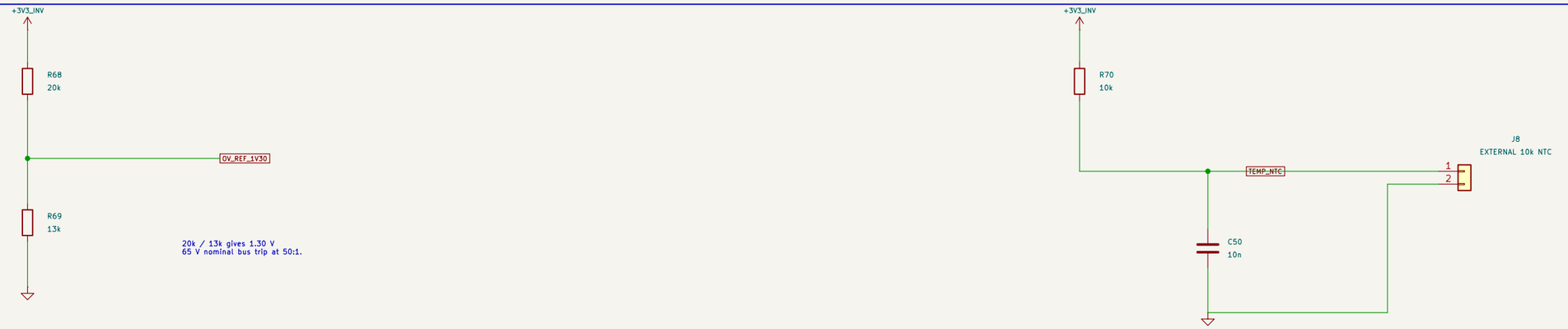
Title: Inverter — three current channels and references

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 9/20

FOUR 50:1 VOLTAGE CHANNELS / COMPLETE DIVIDER TO BUFFER



BUS OVERVOLTAGE REFERENCE / EXTERNAL NTC



Raw phase sense is measured before an external LC filter. Bus RC ~0.98 us; phase RC ~9.8 us.
External NTC nominal 10k; beta, physical placement and cable protection remain to be selected.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

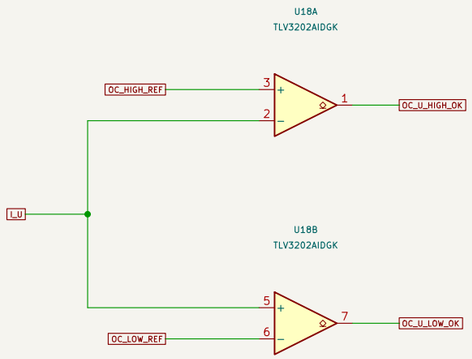
power_system

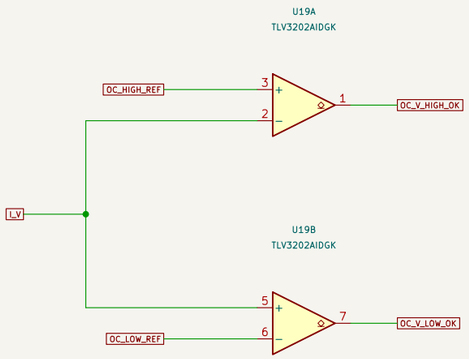
Sheet: /Voltage and NTC/
File: voltage_sense.kicad_sch

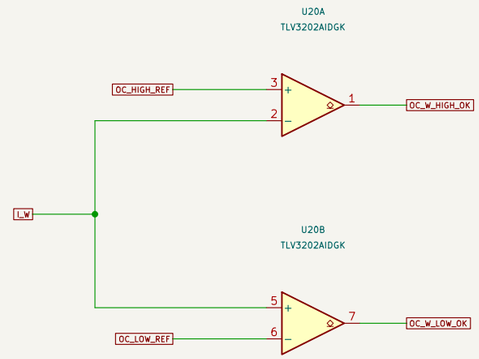
Title: Inverter - voltage and NTC sensing

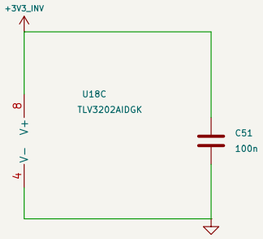
Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 10/20

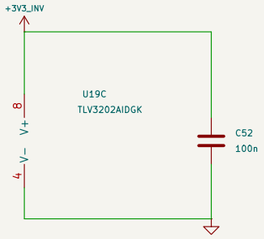
HARDWARE WINDOWS / EACH OUTPUT IS HEALTHY HIGH

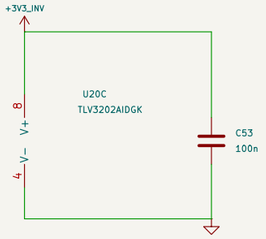




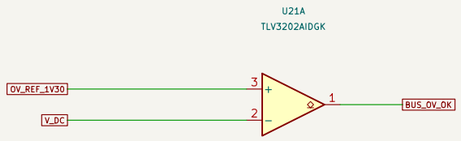


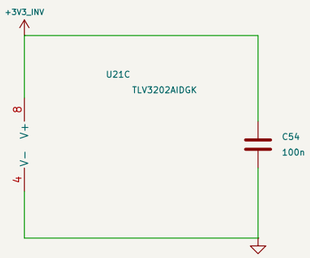


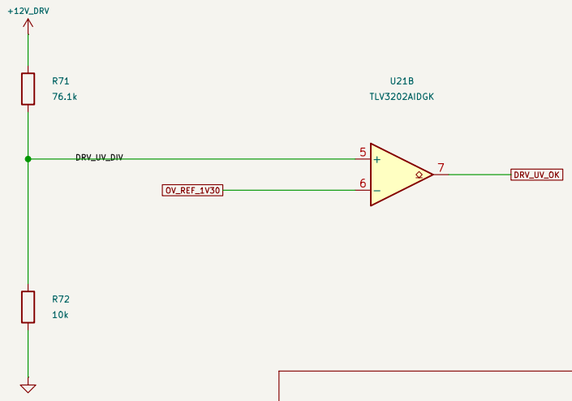




BUS OVERVOLTAGE / DRIVER RAIL UNDERVOLTAGE







Healthy = HIGH. Window violation, bus >~65 V or driver rail <~11.193 V drives its own output LOW.
Push-pull outputs enter separate AND inputs. No wired-AND; no intentional hysteresis; first event is latched.

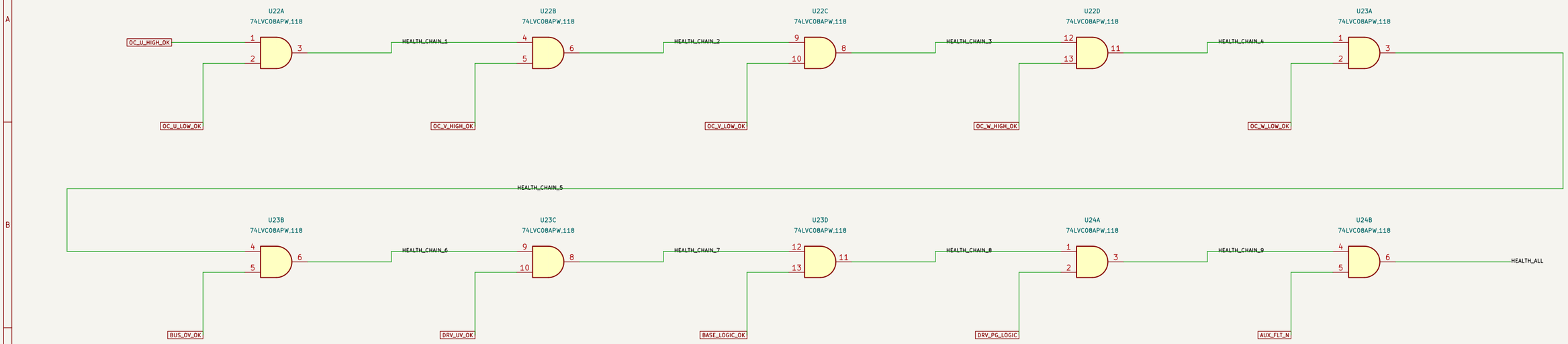
INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Analog protection/
File: comparators.kicad_sch

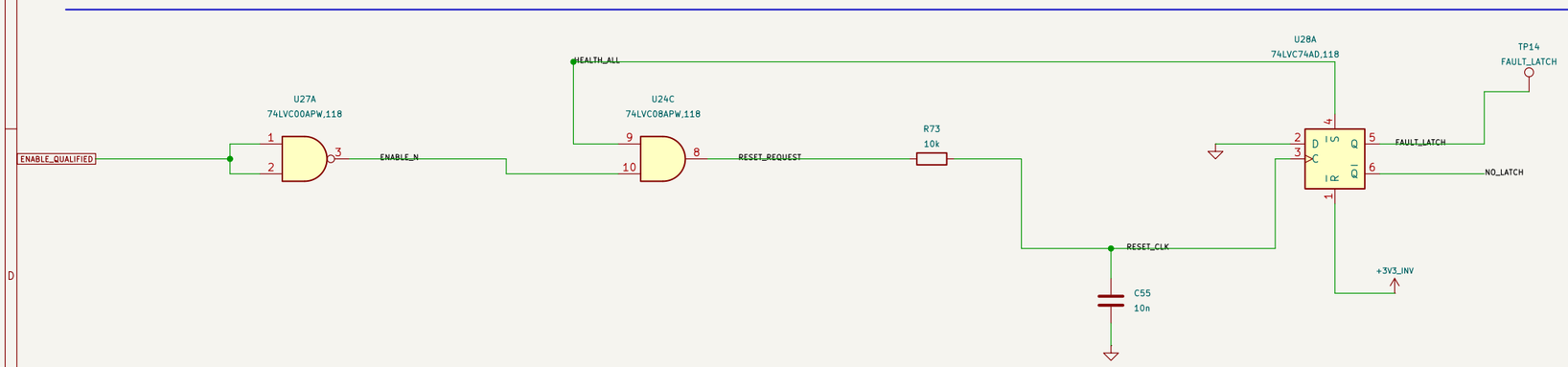
Title: Inverter — current, bus voltage and driver rail comparators

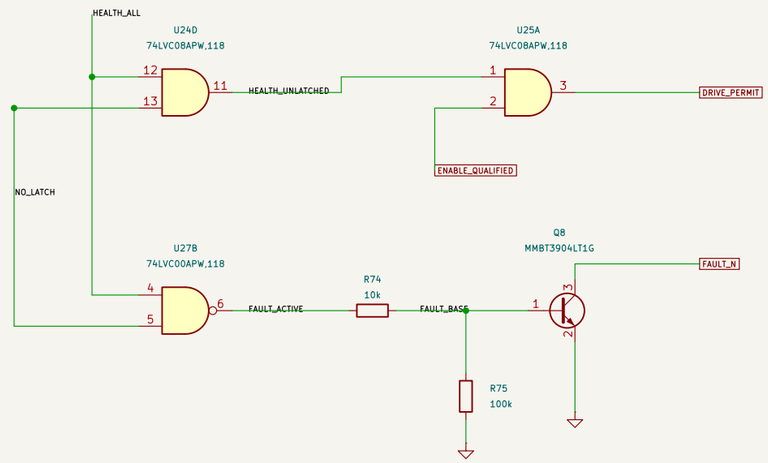
Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 11/20

HARDWARE HEALTH CHAIN / EVERY INPUT MUST REMAIN HIGH



FAULT MEMORY / DELIBERATE DISABLE–TO–RESET / PERMISSION





/S=HEALTH_ALL; /R held HIGH; D=0. Fault asynchronously sets Q.
RESET_REQUEST = HEALTH_ALL AND NOT ENABLE_QUALIFIED.
10k / 10n delays the rising reset clock: hold ENABLE low >=1 ms after recovery.
Fault or invalid supplies block DRIVE_PERMIT directly.
Recovery with ENABLE high cannot clear the latch or restart PWM.
FAULT_N is open collector; core–domain pull–up is on the MCU sub–board.
BASE_READY must be checked separately; an unpowered base cannot assert FAULT_N LOW.
Very short faults during reset may require a fresh disable cycle.

INVERTER | non–isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

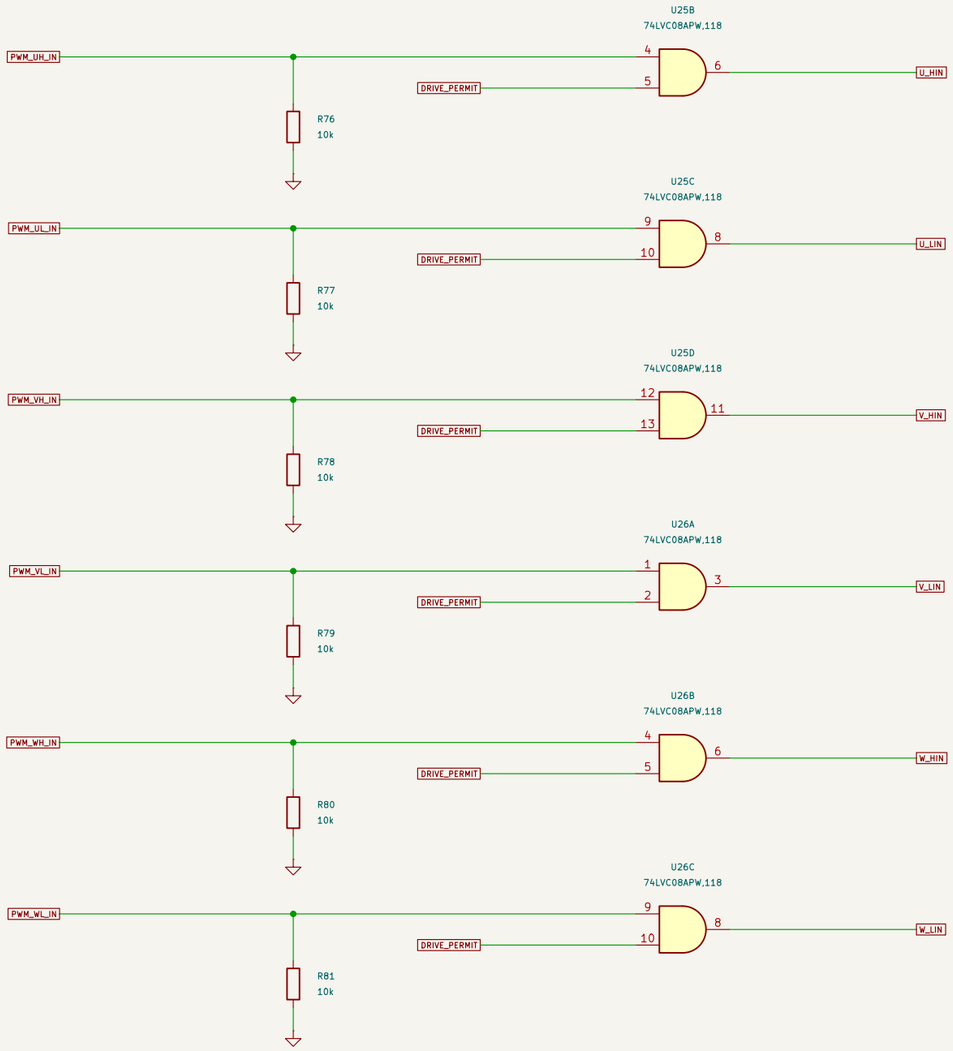
power_system

Sheet: /Hardware latch/
File: fault_logic.kicad_sch

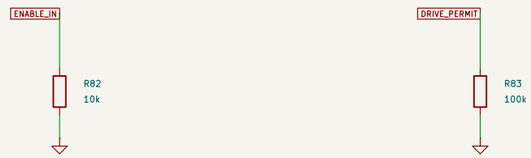
Title: Inverter – fault aggregation and reset latch

Size: A3	Date: 2026–11–07	**Rev: E–core–draft**
KiCad E.D.A. 10.0.7		Id: 12/20

HARDWARE GATING / SIX PWM COMMANDS

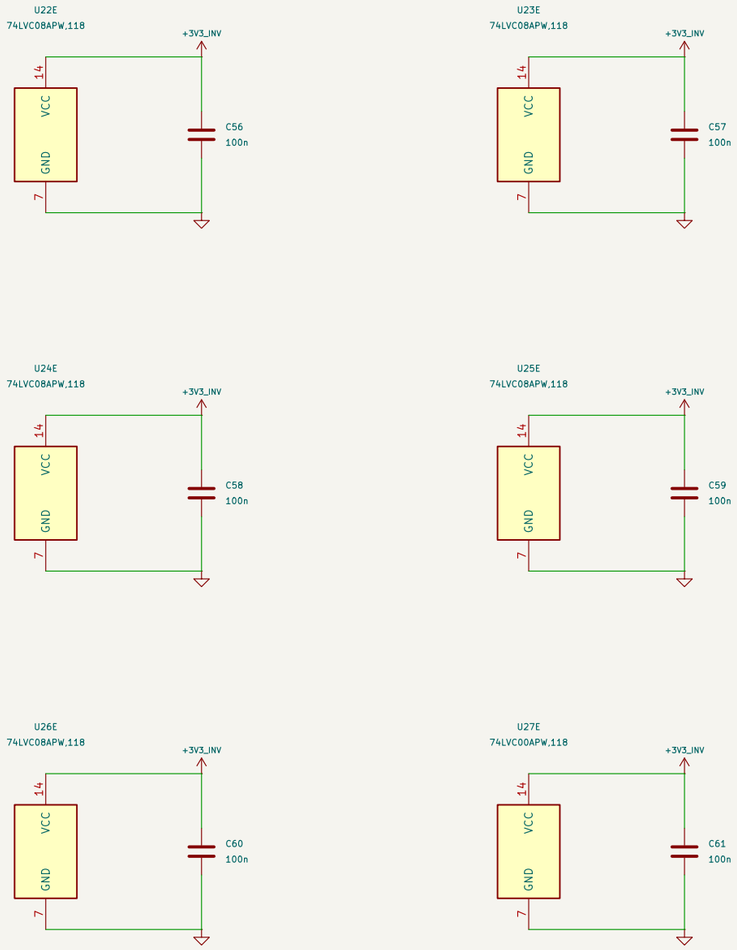


DEFAULT DISABLE / NO PERMISSION



R76–R81: hold all six commands LOW while the controller is disconnected or Hi–Z.
R82: default disabled. R83: default no permission.
Logic levels remain 3.3 V; hardware AND gates operate independently of firmware.

LOGIC POWER / LOCAL DECOUPLING



C56–C61: 100 nF local bypass at each logic package.

U26D / U27C / U27D are used on the Base interlock page.
R76–R81 default PWM LOW; R82 raw ENABLE default LOW.
No MCU on this board; CTRL_READY must qualify ENABLE before any latch reset or DRIVE_PERMIT.

INVERTER | non–isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB
power_system

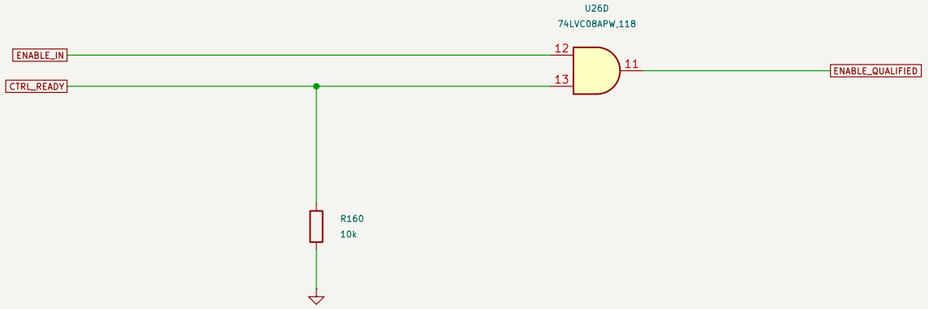
Sheet: /Hardware PWM gates/
File: control_interface.kicad_sch

Title: Inverter – local six PWM hardware gates

Size: A3	Date: 2026–11–07	**Rev: E–core–draft**
KiCad E.D.A. 10.0.7		Id: 13/20

REMOVABLE CONTROLLER / DEFAULT DISABLE

Common GND, 3.3 V logic. CTRL_READY comes from the controller reset-health buffer.
The power base owns all PWM pull-downs and hardware permission; no MCU firmware is required to block a missing controller.



R160 = 10k. Unplugged, controller reset or lost controller power => ENABLE_QUALIFIED LOW.
Only ENABLE_QUALIFIED enters the latch reset and DRIVE_PERMIT logic. Raw ENABLE_IN keeps its original local R82 pull-down.

BASE SUPPLY PRESENCE / INDEPENDENT OF OPEN-COLLECTOR FAULT



BASE_READY = BASE_LOGIC_OK AND DRV_PG_LOGIC. The MCU sub-board provides its receiver pull-down.
FAULT_N remains Q8 open collector with a pull-up only in the MCU 3.3 V domain. A high FAULT_N alone does not prove base power.
U26/U27 package power and 100 nF bypass stay on Hardware PWM gates. No added logic package.

Existing Rev C PCB is historical and does not match Rev D
POWER BASE | COMMON GND | SCHEMATIC REVIEW DRAFT
power_system

Sheet: /Base interlock/
File: base_interlock.kicad_sch

Title: Power base — controller readiness and local supply interlock

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 14/20

LOCAL POWER INDICATOR / NO MCU LED SIGNALS ACROSS HEADER



D3: base 3.3 V present. This lamp alone does not indicate driver power or protection permission.
D4..D7 and R131..R134 belong to the removable MCU sub-board: RUN / FAULT / COMM / AUX.
The power base has no MCU-run or communication LED nets on its board-to-board header.

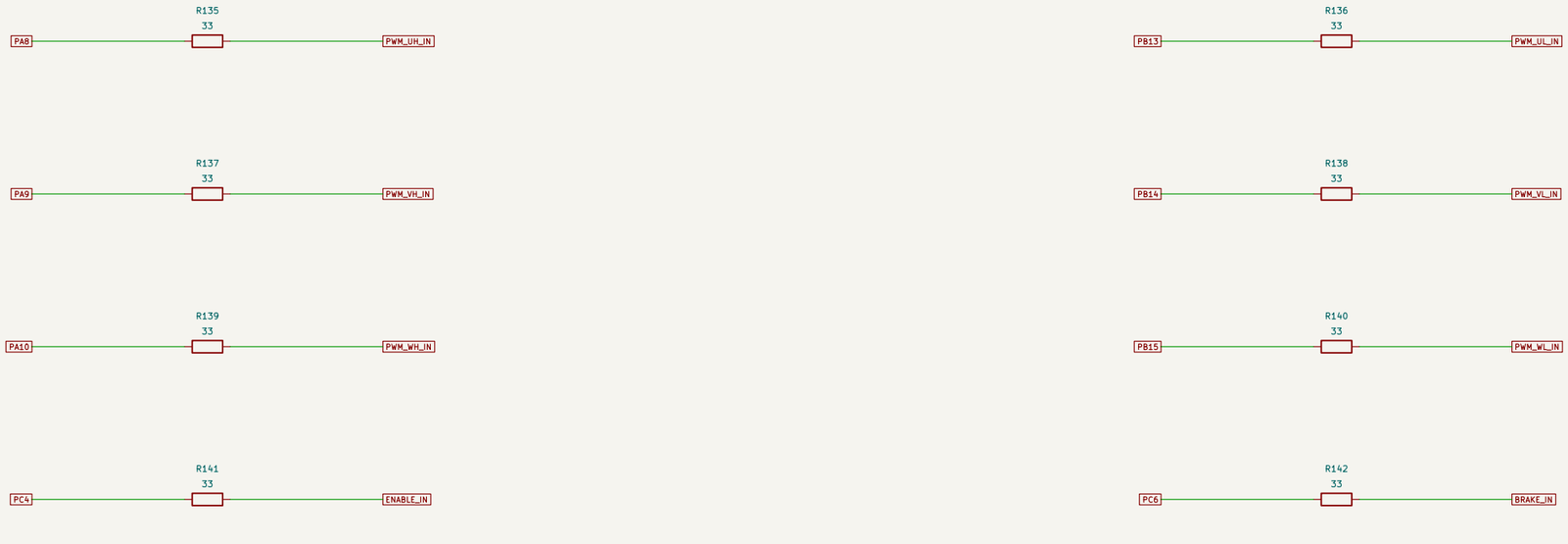
Existing Rev C PCB is historical and does not match Rev D
POWER BASE | COMMON GND | SCHEMATIC REVIEW DRAFT
power_system

Sheet: /Base LEDs/
File: base_leds.kicad_sch

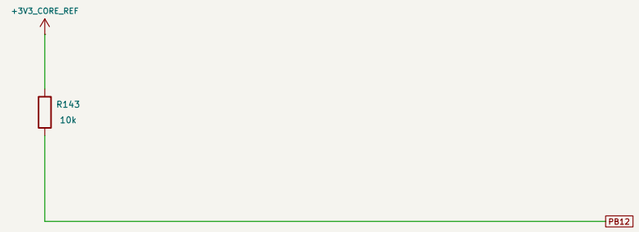
Title: Power base – local power indication

Size: A4	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 15/20

CONTROLLER COMMAND SERIES RESISTORS / BASEBOARD SIGNALS



FAULT INPUT / LOCAL PULL-UP AND TIMER BKIN



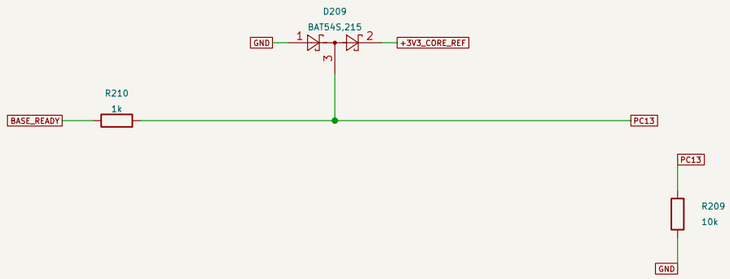
Protected PB12 is TIM1_BKIN: pull-up uses the exported core 3V3 reference.
NRST is debug-only: it is not a power-stage fault output.
Six PWM, ENABLE and BRAKE series resistors remain local to the base.

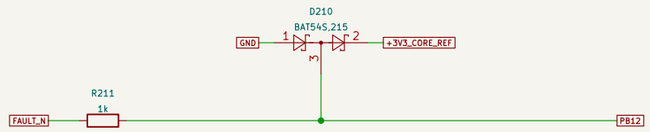
CONTROLLER | STM32G431RBT6 | shared-ground replaceable MCU daughterboard
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system
Sheet: /GPIO command coupling/
File: mcu_coupling.kicad_sch

Title: Baseboard — GPIO command coupling and fault pull-up

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 16/20

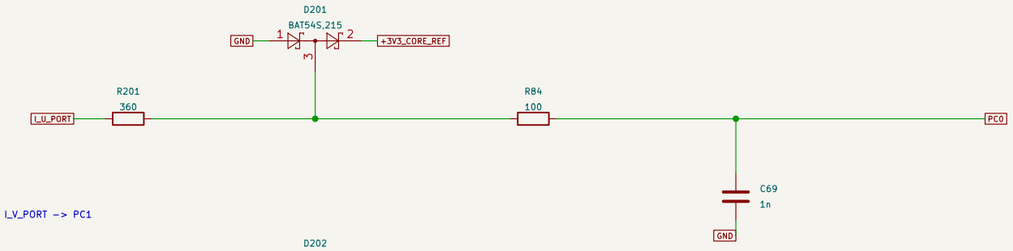
STATUS RETURN LIMITING / CORE-REFERENCED CLAMPS



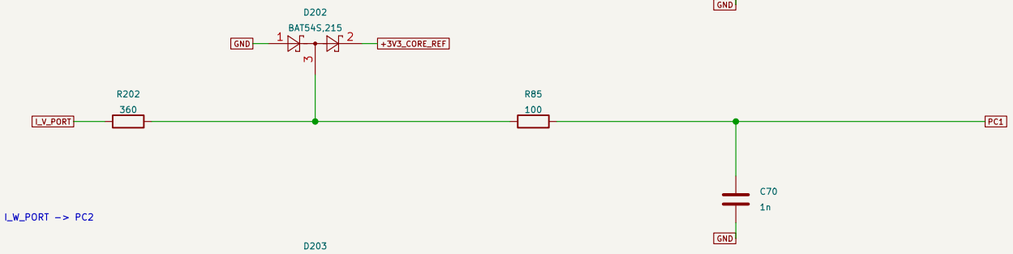


8 ADC PATHS / 360R CLAMP + 100R / 1n AT CORE SOCKET

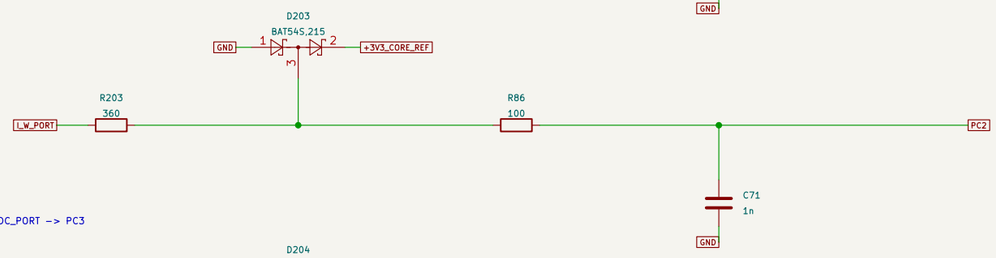
I_U_PORT -> PC0



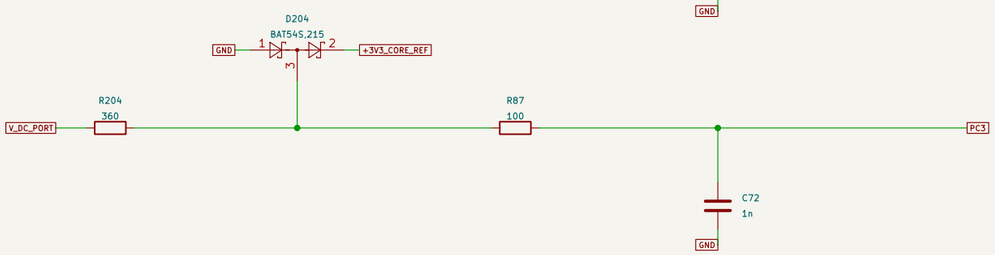
I_V_PORT -> PC1



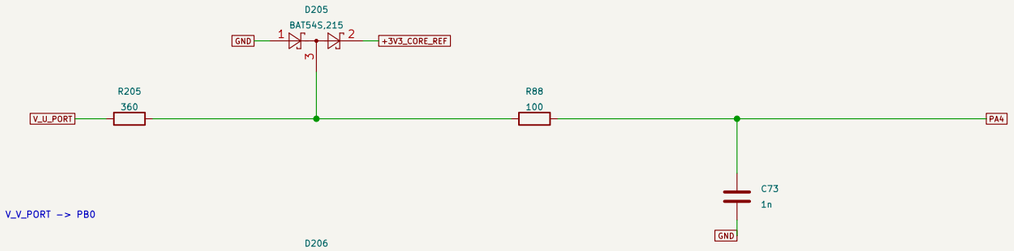
I_W_PORT -> PC2



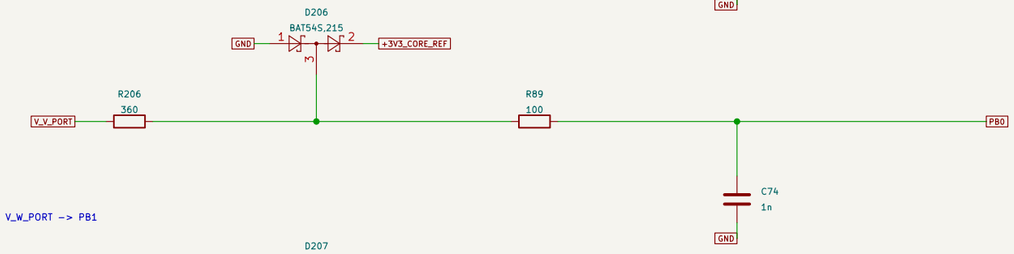
V_DC_PORT -> PC3



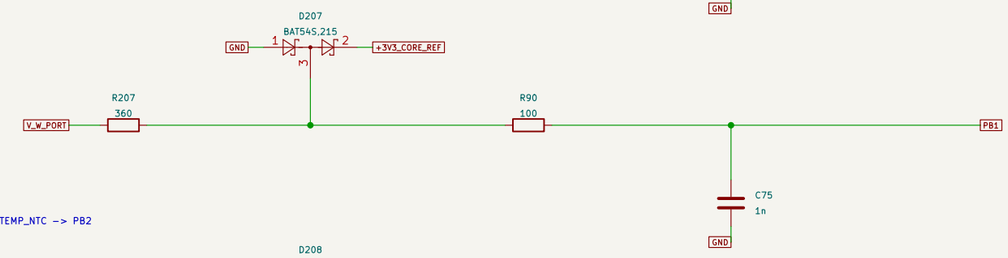
V_U_PORT -> PA4



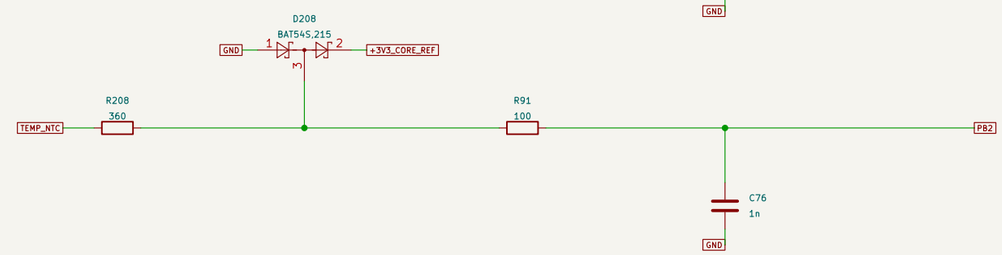
V_V_PORT -> PB0



V_W_PORT -> PB1



TEMP_NTC -> PB2



Clamp upper rail returns ONLY through core 3V3_REF pins; never join it to base +3V3_INV. Power off both boards before mating.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB

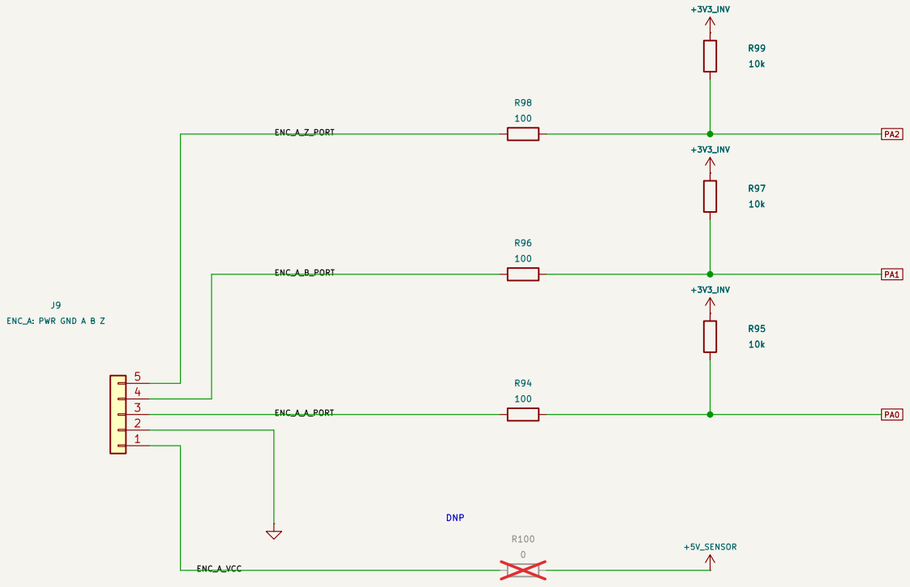
power_system

Sheet: /ADC and status returns/
File: core_signal_protection.kicad_sch

Title: Baseboard — ADC and status return conditioning

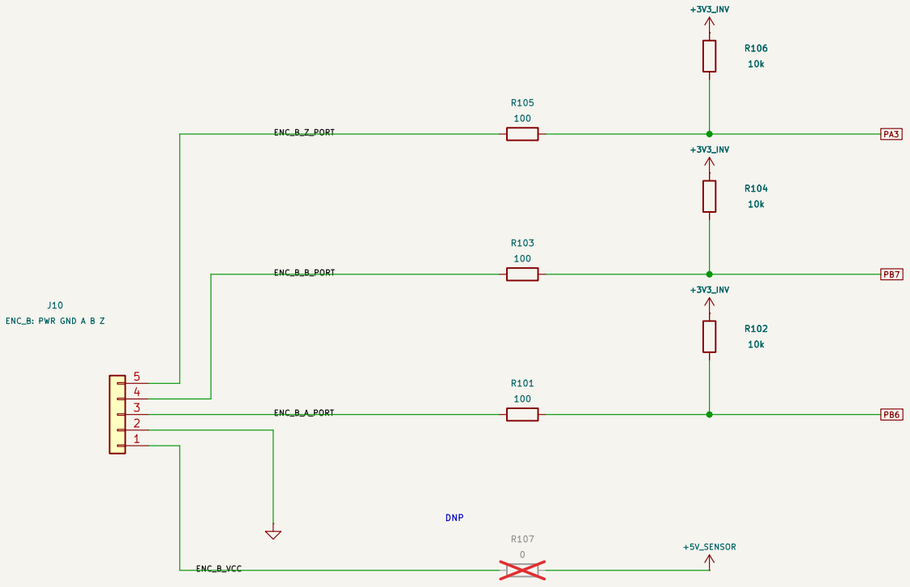
Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 17/20

ABZ ENCODER A / 3.3 V SINGLE-ENDED



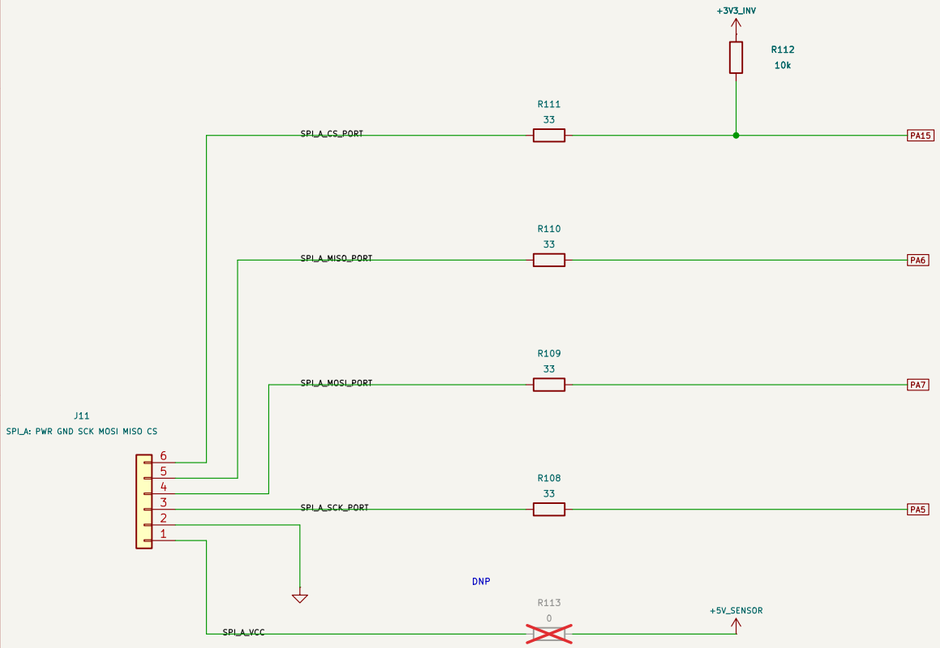
100 ohm series / 10k pull-ups to 3.3 V. 5 V supply link DNP.

ABZ ENCODER B / 3.3 V SINGLE-ENDED

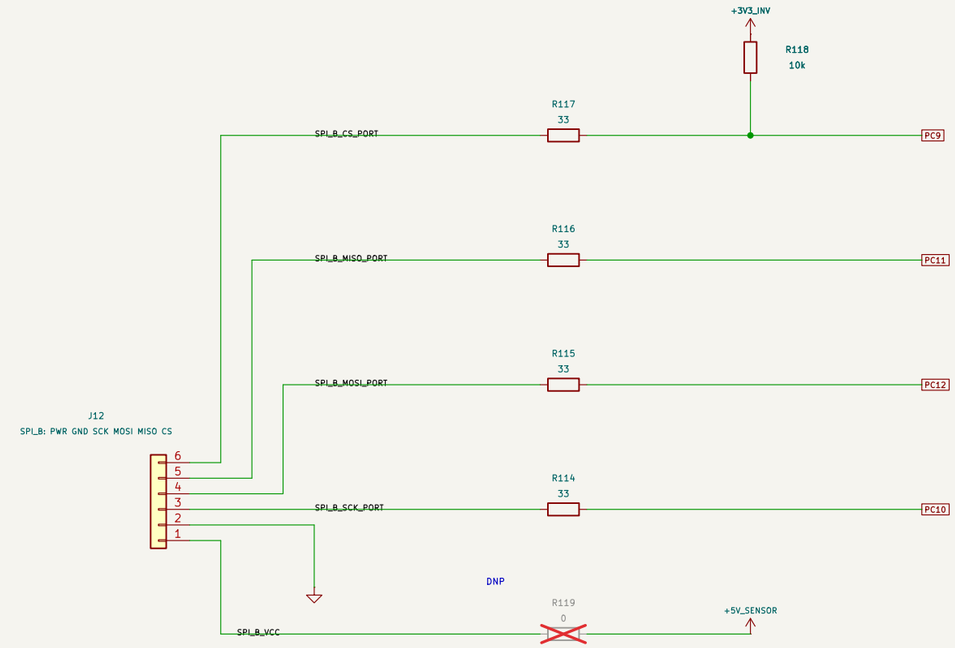


100 ohm series / 10k pull-ups to 3.3 V. 5 V supply link DNP.

SPI SENSOR A / 3.3 V SINGLE-ENDED



SPI SENSOR B / 3.3 V SINGLE-ENDED



All external signal pins are 3.3 V single-ended; no 5 V level conversion or cable protection is claimed.
R100 / R107 / R113 / R119 are DNP: external sensor-power links remain open by default.

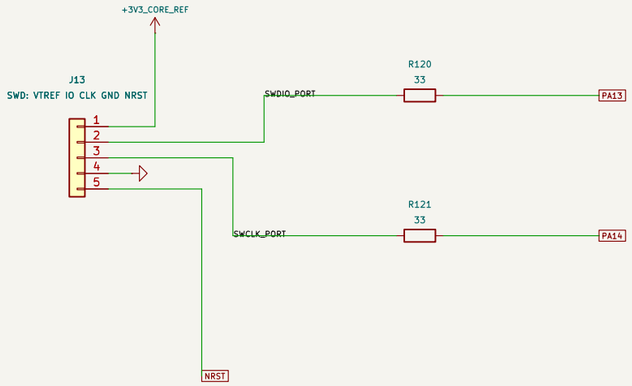
CONTROLLER | STM32G431RBT6 | shared-ground replaceable MCU daughterboard
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Encoders and SPI/
File: sensors.kicad_sch

Title: Baseboard — encoders and SPI

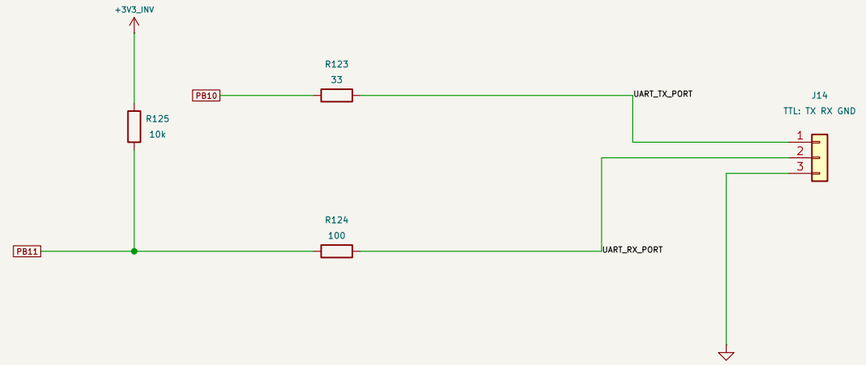
Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 18/20

SWD / OPEN-DRAIN RESET / BOOT



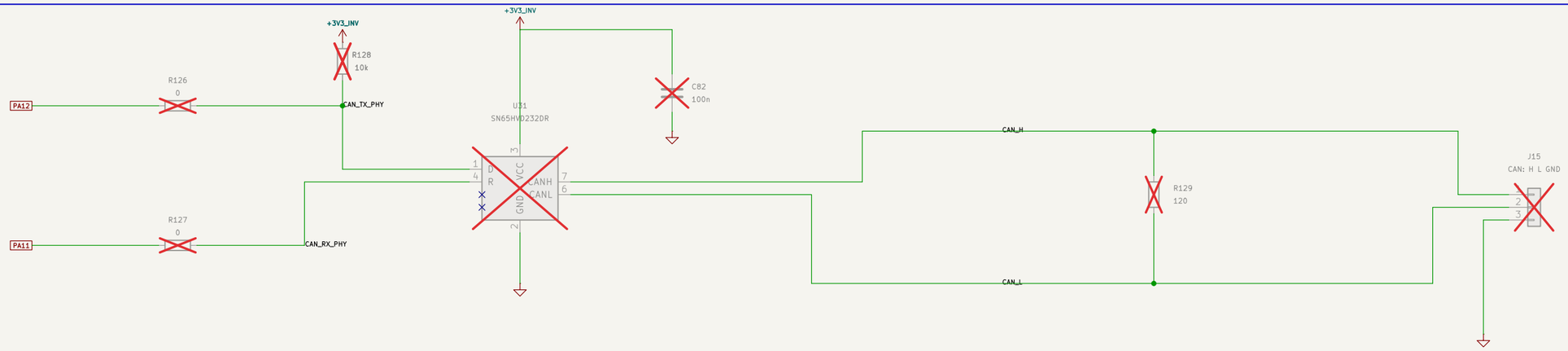
VTREF senses the core 3V3 reference. NRST goes only to the core reset domain.
RESET / BOOT buttons are on the minimum-system core.
Existing TTL PB10/PB11 is application UART, not the ROM USART3 pin pair.

TTL UART / USART3 / 3.3 V

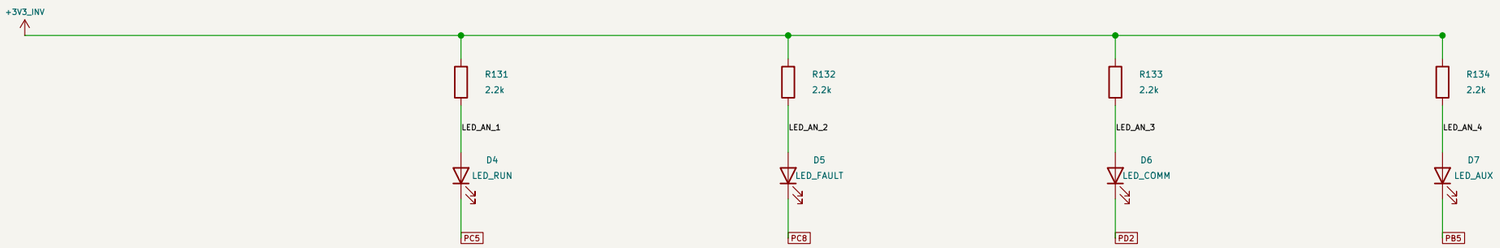


TX/RX relative to this board: no RS-232 levels.
No USB and no supply output on TTL connector.

OPTIONAL CLASSIC CAN / COMPLETE PHY BLOCK DNP



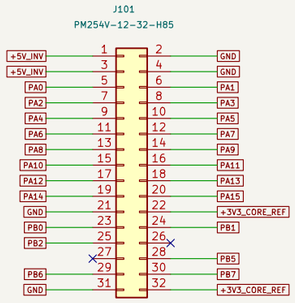
FOUR MCU STATUS INDICATORS / ACTIVE LOW



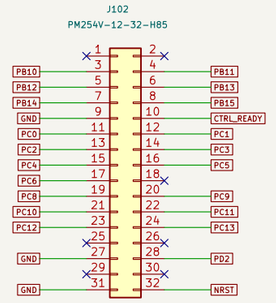
CONTROLLER | STM32G431RBT6 | shared-ground replaceable MCU daughterboard
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /SWD UART CAN and status/		
File: debug_comms.kicad_sch		
Title: Baseboard — SWD TTL optional CAN and status LEDs		
Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 19/20

J101 / PA0–15 + PB0–7



J102 / PB8–15 + PC0–15 + PD/PF/NRST



64 contacts: 52 MCU signal pads + 2 core 5V feed + 2 core 3V3 reference + CTRL_READY + 7 GND.

All MCU GPIOs are exposed by port name. PG10 stays NRST; PF0/PF1 share optional HSE; PB8 shares BOOT.

PC7 is RUN_ARM when plugged into this power base. CTRL_READY = Schmitt-reset-ok AND PC7; default LOW.

Standalone core: regulated 4.75–5.25V at J101 pins1/3 and GND. Do not parallel independent 5V sources.

3V3 reference is core -> base clamps / fault pull-up / SWD sense only. Never parallel base and core regulators.

Application ABZ / SPI / UART / SWD / CAN / LEDs / ADC filters all reside on the power baseboard.

60x60 POWER ABOVE / 60x60 CORE BELOW. Core TOP male -> power BOTTOM socket. Board gap / thermal / button access pending PCB.







Nine unused socket contacts are NC on this base only; all corresponding GPIO pads remain exposed on the core.

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

power_system

Sheet: /64-pin core sockets/
File: power_interface.kicad_sch

Title: Baseboard minimum-core sockets

Size: A3	Date: 2026-11-07	**Rev: E-core-draft**
KiCad E.D.A. 10.0.7		Id: 20/20